IEEE
AI Team : Task 3

Thesis
Titanic Data Findings

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Loading The Data

First, we starting by loading the CSV file into the editor as a Pandas data frame.

After so, we take a quick look on the data to determine which fields got corrupted (NaN) values and how many of them to have an insight before we start data cleaning.

```
raw_data=pd.read_csv("titanic.csv",sep=",")
print(raw_data.info())
print(raw_data.head())
```

Data cleaning

After loading the data, we quickly notice that most of the data was intact except for the age and cabin columns which had the most amount of NaNs, with Age having about 20% of its fields being NaNs and Cabin having around 78% of its fields being NaNs.

Since its intuitive to think that age would be of great importance for survival and that its not as corrupted, I decided to fill the NaNs of age with the age mean value.

I also decided to drop the cabin column as its immensely corrupted and wouldnt be of use for the data.

We also made sure that the “Sex” column only had male or female for further analysis.

```
# 2) cleaning data:
#    check where it is NULL
print(raw_data.isnull().sum())
#    we see that there are 177 nulls and 687 nulls in cabin
#    so there's about 20% of ages that are null and 78% of cabins that are null
#    Since age is an intuitevly important factor in survival and not much is missing
#    I will fill the nulls of age with mean value but drop the cabin data column since
#    most of it is missing
raw_data["Age"] = raw_data["Age"].fillna(raw_data["Age"].mean())
df = raw_data.copy() # to copy the data before anymore changes
df = df.drop(columns=["Cabin"])
print(df.info())
print(df.isnull().sum())
#    we see now that the age column has no nulls and the cabin column has dropped
```

Exploratory Data Analysis (EDA)

We perform a couple of operations on data to get useful insights about the titanic and survivorship in general.

Relation between classes and survival rate

1) We notice that first class had a survival rate of 62% with 133 people surviving.
2) Second class had a survival rate of 44% with 82 people surviving.
3) Third class had a survival rate of 24% with 117 people surviving.

It appears that the position of cabins based on classes with first class in the middle of the ship to reduce motion and such helped with survival and less impact of the collision on people.

Relation between number of males and females

1) We notice that out of survivors, males made up only 33%, while females made up 67% of the survivors.

So its safe to assume that women cabins were in places that were more secure and afloat than men.

Relation between age of males and females

1) The mean age of men was 30.5.
2) The mean age of women was 30.5.
3) The mean age of men who survived was 27.6
4) The mean age of women who survied was 27.9.

```
# 3) Exploratory Data Analysis (EDA)
#    First we check which classes survived the most as different classes often have
#    different postions on the ship like the front or the back and so on
#print(df[["Survived", "Pclass"]])
print((df["Pclass"]==1).sum() )
#    we have around 216 people in first class
print(df[(df["Survived"]>0)& (df["Pclass"]==1)].sum()) # 133
#    out of 216 in first class, 133 people survived which is about 62% survival rate
print((df["Pclass"]==2).sum() ) # 184
print(df[(df["Survived"]>0)& (df["Pclass"]==2)].sum()) # 82
#   out of 184 in second class, 82 people survied which is 44% survival rate
print((df["Pclass"]==3).sum() ) # 491
print(df[(df["Survived"]>0)& (df["Pclass"]==3)].sum()) #117
#   out of 491 in third class, only 119 people survived which is about 24% survival rat
#   as we can see that being higher class, thus being in the middle of the ship when th
#   when the crash happened, helped much
#  next we look at mean the of the ages that survived in men and women
print(df[(df["Sex"]=="male")& (df["Survived"]==1)].sum()) # 109 male survied
print(df[(df["Sex"]=="female")& (df["Survived"]==1)].sum()) #223
#   The total number of males survied is 109 out of 332 survivors
```

```
#  next we look at mean the of the ages that survived in men and women
print(df[(df["Sex"]=="male")& (df["Survived"]==1)].sum()) # 109 male survied
print(df[(df["Sex"]=="female")& (df["Survived"]==1)].sum()) #223
#   The total number of males survied is 109 out of 332 survivors
#   The total number of males survied is 109 out of 332 survivors
mean_male_age = df[(df["Sex"] == "male")]["Age"].mean()
print(mean_male_age)
mean_female_age = df[(df["Sex"] == "male")]["Age"].mean()
print(mean_female_age)
mean_age_male_survivors = df[(df["Sex"] == "male") & (df["Survived"] == 1)]["Age"].mean
print(mean_age_male_survivors)
mean_age_male_survivors = df[(df["Sex"] == "female") & (df["Survived"] == 1)]["Age"].me
print(mean_age_male_survivors)
#   as we can see that even tho the number of survivors is different between sexes but
#   the average age that survived reamained young
#-------------------------------------------------------
#  4) Visualisations
survivors_per_class = df.groupby("Pclass")["Survived"].sum()
survivors_per_class.plot(kind="bar", color="#981012")
```

Visualisations

```
#  4) Visualisations
survivors_per_class = df.groupby("Pclass")["Survived"].sum()
survivors_per_class.plot(kind="bar", color="#981012")

plt.title("Total Survivors per Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Survivors")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

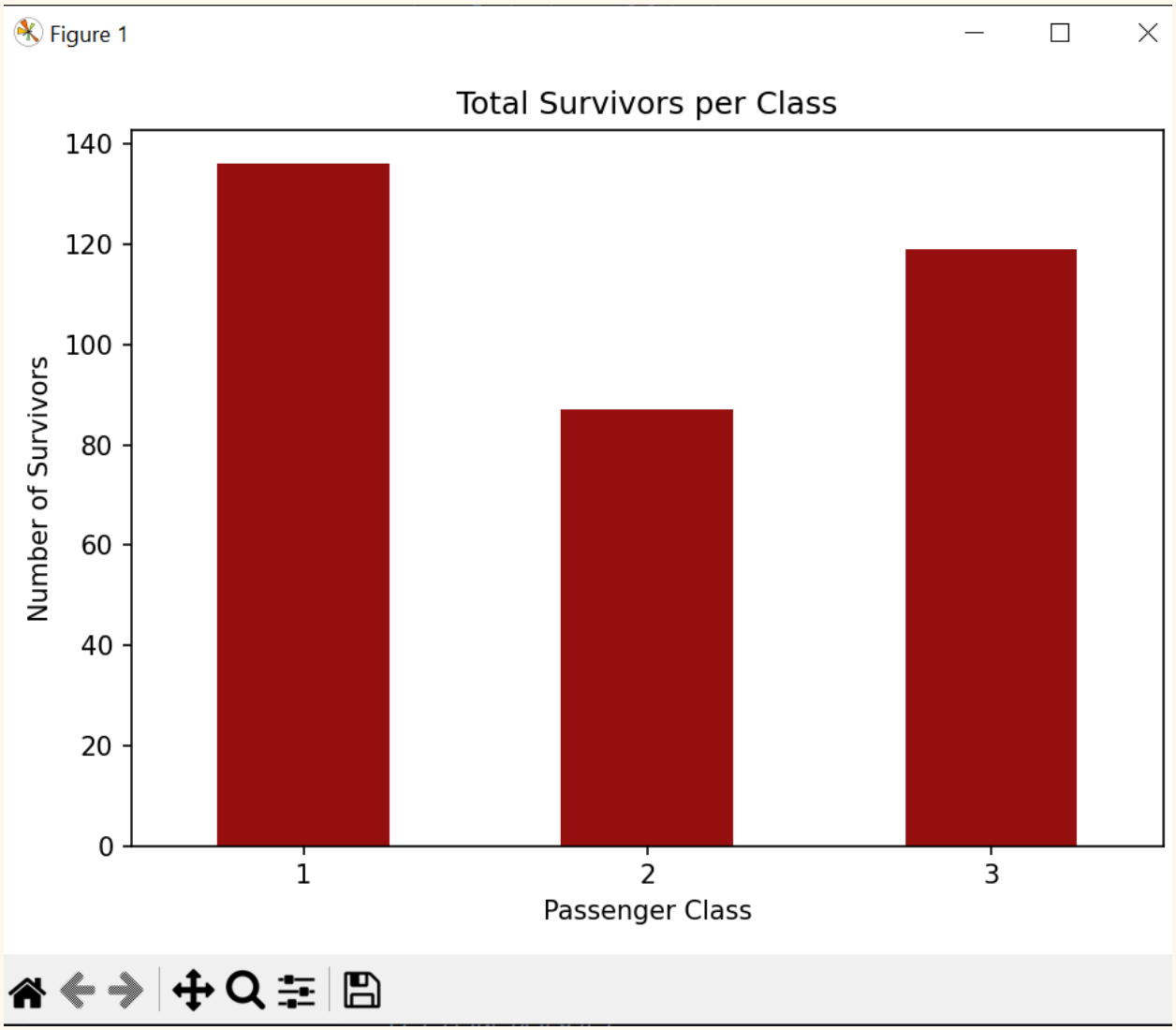
Figure 1
Total Survivors per Class
Number of Survivors
0
20
40
60
80
100
120
140
1
2
3
Passenger Class